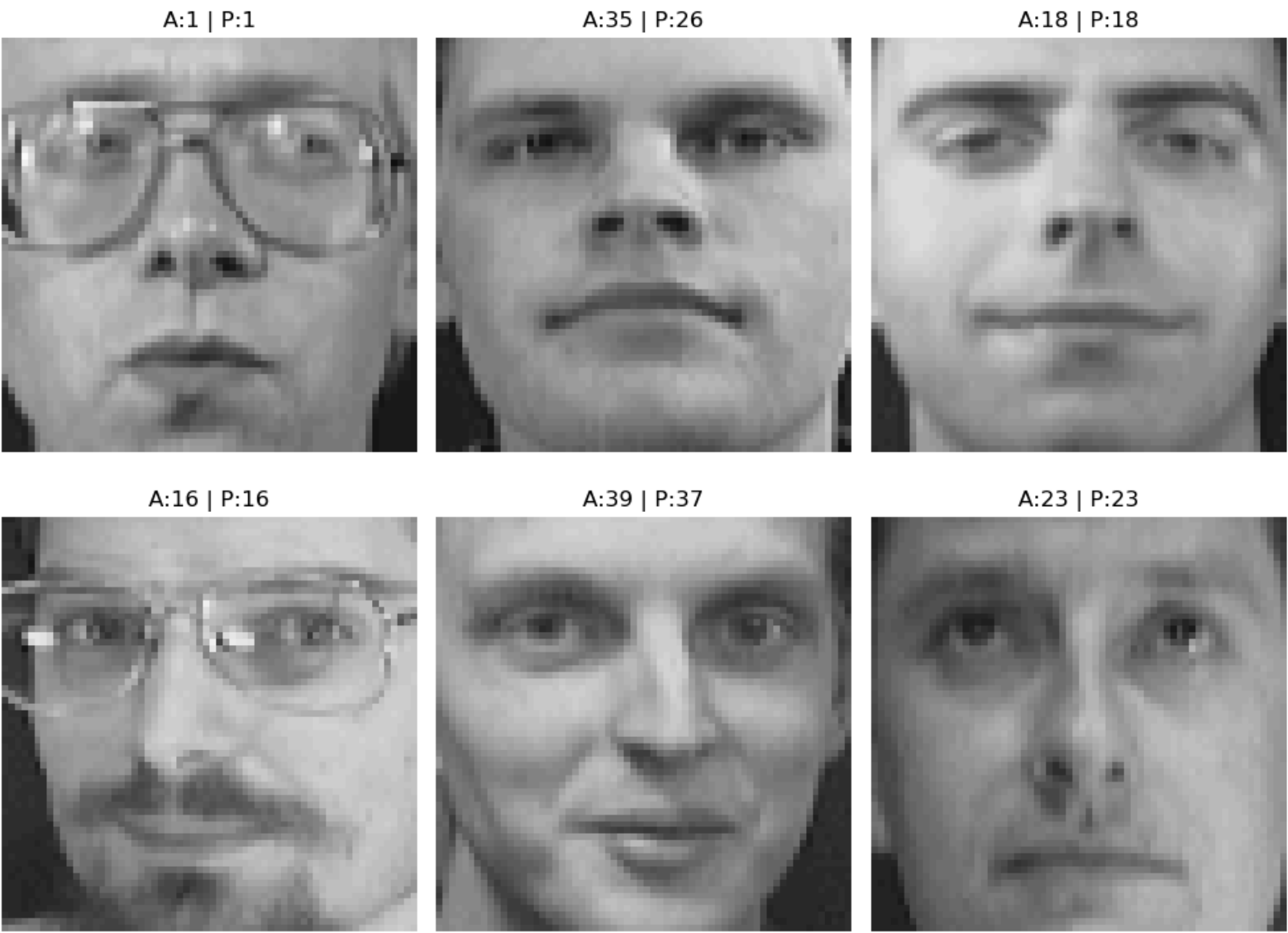


```
In [1]: import numpy as np
import cv2
import os
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_olivetti_faces
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
path = r"C:\Users\Dhanashri\Downloads\Images\cifar-10-batches-py"
try:
    if os.path.exists(path) and path.endswith(".npy"):
        imgs = np.load(path)
        labels = np.array([i // 10 for i in range(len(imgs))])
    else:
        data = fetch_olivetti_faces()
        imgs = data.images
        labels = data.target
except Exception as e:
    print("Dataset loading error:", e)
    data = fetch_olivetti_faces()
    imgs = data.images
    labels = data.target
print("Dataset Shape:", imgs.shape)
print("Classes:", len(np.unique(labels)))
color_imgs = []
for img in imgs:
    img = (img * 255).astype("uint8")
    if len(img.shape) == 2:
        img = cv2.cvtColor(img, cv2.COLOR_GRAY2BGR)
    color_imgs.append(img)
color_imgs = np.array(color_imgs)
features = []
for im in color_imgs:
    hist_features = []
    for c in range(3):
        hist = cv2.calcHist([im], [c], None, [256], [0, 256])
        hist = cv2.normalize(hist, None).flatten()
        hist_features.extend(hist)
    features.append(hist_features)
features = np.array(features)
print("Feature Shape:", features.shape)
Xtr, Xte, Ytr, Yte, Imtr, Imte = train_test_split(
    features,
    labels,
    color_imgs,
    test_size=0.2,
    random_state=42,
    stratify=labels
)
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
model.fit(Xtr, Ytr)
pred = model.predict(Xte)
acc = accuracy_score(Yte, pred)
print("Accuracy:", round(acc * 100, 2), "%")
plt.figure(figsize=(10, 8))
for i in range(6):
    plt.subplot(2, 3, i + 1)
    img_rgb = cv2.cvtColor(Imte[i], cv2.COLOR_BGR2RGB)
    plt.imshow(img_rgb)
    plt.title(f"A:{Yte[i]} | P:{pred[i]}")
    plt.axis("off")
plt.suptitle("Face Recognition Output")
plt.tight_layout()
plt.show()
```

Dataset Shape: (400, 64, 64)
Classes: 40
Feature Shape: (400, 768)
Accuracy: 78.75 %

Face Recognition Output



In []: